

Supplementary material for ‘Latent coverage of noisy calibration thresholds’

Kun Woo Park

Equation, theorem and section numbers without the prefix S refer to the main paper. Numbers that are not proved are labelled as numerical, and the precision of each computation is stated.

S1 Proofs omitted from the main paper

Proof of Proposition 1. Since $\mathcal{F} \subset \mathcal{L}$ it suffices to bound the supremum over $\mathcal{L}$. Let $W \in \mathcal{L}$ satisfy $E\{g_x(W)\} \geq p$ and let $s < Q_q(|W|)$, so that $\text{pr}(|W| < s) \leq q - \gamma$ for some $\gamma > 0$. Choose $M \geq s$ with $g_x(w) \leq p$ for $|w| \geq M$, and let W_M be W conditioned on $|W| \leq M$, which is log-concave. With $\varepsilon_M = \text{pr}(|W| > M)$, the conditioning removes only mass on which the kernel is at most p , so

$$E\{g_x(W_M)\} \geq \frac{p - p\varepsilon_M}{1 - \varepsilon_M} = p, \quad \text{pr}(|W_M| < s) \leq \frac{q - \gamma}{1 - \varepsilon_M} < q$$

for M large. On $[-M, M]$ the constraint function $g_x - p$ is continuous, and $\mu \mapsto -\mu(-s, s)$ is linear and upper semicontinuous for weak convergence because $(-s, s)$ is open. By the localization theorem (Kannan et al., 1995; Fradelizi & Guédon, 2004), the supremum of this functional over the log-concave probability measures on $[-M, M]$ that satisfy $\int (g_x - p) d\mu \geq 0$ is attained at a point mass or at a law with log-affine density on a segment. Hence some member of $\mathcal{F}$ is feasible and has $\text{pr}(|W| < s) < q$, that is, $Q_q(|W|) \geq s$. Since $s < Q_q(|W|)$ was arbitrary the claim follows. Only continuity of the kernel and its decay at infinity were used, so the argument applies to the average kernel $\bar{g}$ of § 4.2. $\square$

For $\beta > 0$ and $\sigma = x^{1/2}$, integration by parts gives

$$\int_a^b e^{\beta w} \Phi\left(\frac{c - w}{\sigma}\right) dw = \frac{1}{\beta} \left[e^{\beta w} \Phi\left(\frac{c - w}{\sigma}\right) \right]_a^b + \frac{e^{\beta c + \beta^2 \sigma^2 / 2}}{\beta} \left\{ \Phi\left(\frac{b - c - \beta \sigma^2}{\sigma}\right) - \Phi\left(\frac{a - c - \beta \sigma^2}{\sigma}\right) \right\}, \quad (\text{S1})$$

so the constraint is a combination of normal distribution functions. For fixed slope and length, the translations c of a member of $\mathcal{F}$ that satisfy the constraint form an interval, because $c \mapsto \text{pr}(|W + c + e| \leq 1)$ is log-concave (Prékopa, 1973), and $c \mapsto \text{pr}(|W + c| \leq s)$ is log-concave, so $c \mapsto Q_q(|W + c|)$ is quasi-convex and is maximized at an end of that interval. Each end is found by a root-finding in c , and the quantile by a second root-finding, using (S1). This two-dimensional computation over slope and length, and an independent differential-evolution search over the three-parameter family, agree to 1.5×10^{-13} at all 73 tabulated values of $x \geq 0.006$ (numerical).

Proof of Lemma 1. Exponential tails are log-concave, which gives one inequality. Conversely, let $Y \in \mathcal{L}$ with $\text{pr}(Y + Z \leq 0) \geq p$. If $Y = y_0$ is a point mass then $y_0 \leq -\Phi^{-1}(p) = \lim_{u \rightarrow \infty} \{b_u(p) - u^{-1} \log(1/q)\}$. Otherwise $F = F_Y$ is continuous and $\log F$ is concave (Prékopa, 1973). Let $v = F^{-1}(q)$ and $G(y) = F(v + y)$. The point 0 is interior to $\{G > 0\}$, so the right derivative $u \geq 0$ of $\log G$ at 0 is a supergradient, and $G(y) \leq qe^{uy}$ for all y . If $u = 0$ then $G \leq q < 1$, which is impossible, so $u > 0$ and $G(y) \leq \min(qe^{uy}, 1)$,

which is the distribution function of $b_0 - E_u$ with $b_0 = u^{-1} \log(1/q)$. Hence $Y' = v + b_0 - E_u$ lies stochastically below Y and has q -quantile v , and

$$p \leq \Pr(Y + Z \leq 0) \leq \Pr(Y' + Z \leq 0) = m_u(v + b_0).$$

Since m_u is decreasing, $v \leq b_u(p) - b_0$. For monotonicity, let $q' > q$, let Y be feasible at (q', q') with $v = F_Y^{-1}(q')$, and put $\rho = (q' - q)/(1 - q)$. Then Y conditioned on $Y \geq F_Y^{-1}(\rho)$ is log-concave; over $\mathcal{B}$ it stays in $\mathcal{B}$, because $\log(F - \rho) = h(\log F)$ with $h(z) = \log(e^z - \rho)$ nondecreasing and concave, and the log survival function of the conditioned law is $\min\{0, \log(1 - F) - \log(1 - \rho)\}$, a minimum of concave functions. It satisfies the constraint at level $(q' - \rho)/(1 - \rho) = q$, because the removed mass has noisy constraint value at most one, and has q -quantile v . Point masses have value $-\Phi^{-1}(q') < -\Phi^{-1}(q) \leq c_q$. Finally, $c_q > 0$ because $v_q(u) = u^{-1} \log\{e^{u^2/2}\Phi(b_u - u) + e^{ub_u}\Phi(-b_u)\} = u/2 + o(u)$ as $u \rightarrow 0$, where $b_u = b_u(q)$ and $ub_u \rightarrow \log(1/q)$. $\square$

Proof of Theorem 1, lower bound. Fix $\epsilon > 0$ and choose u with $v_q(u) > c_q - \epsilon$. For $p' \in [q, (1 + q)/2]$, $b_u(p') \geq b_{\min} > -\infty$. Let $Y = b_u(p') - E_u$ and $W = 1 + x^{1/2}Y$. With $\tau = 2x^{-1/2}$,

$$\Pr(|W + x^{1/2}Z| \leq 1) = p' - \Pr(Y + Z < -\tau) \geq p' - \eta_x, \quad \eta_x = e^{-u(\tau + b_{\min})/2} + \Phi\{-(\tau + b_{\min})/2\},$$

by splitting the event $Y + Z < -\tau$ according to whether $Z < -(\tau + b_{\min})/2$. Take $p' = q + \eta_x$, which lies in $[q, (1 + q)/2]$ for small x . Since $\Pr(|W| \leq s) \leq F_W(s)$, $Q_q(|W|) \geq 1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1} \log(1/q)\}$. As $\eta_x \rightarrow 0$ and b_u is continuous, $\liminf_{x \rightarrow 0} \{R_{q,q}(x) - 1\}/x^{1/2} \geq c_q - \epsilon$. $\square$

Proof of Lemma 2. For $q > 1/2$, v_q is continuous on $(0, \infty)$, tends to 0 as $u \rightarrow 0$ and to $-\Phi^{-1}(q) < 0$ as $u \rightarrow \infty$, and is positive somewhere, so its maximizers form a nonempty compact subset of $(0, \infty)$ and κ_q^* is well defined. Let $a = \log(1/q)$ and $A = 1 - a > 0$. Let Y be nondegenerate with q -quantile v . Since $\log F_Y$ is concave, a supergradient $u > 0$ at v gives $F_Y(y) \leq \min\{qe^{u(y-v)}, 1\}$, the distribution function of $Y' = v + a/u - E_u$; so Y' lies stochastically below Y and has the same q -quantile. Then $E(Y') \leq E(Y) \leq -\kappa$ and $\Pr(Y' + Z \leq 0) \geq q$, so the replacement keeps both constraints. For the tail $b - E_u$ the constraints read $b \leq b_u(q)$ and $b \leq u^{-1} - \kappa$, so the largest quantile is $\min\{v_q(u), A/u - \kappa\}$. If $k_q(u) < \kappa$ this minimum is $A/u - \kappa$; since k_q is continuous and $k_q(s) \rightarrow \infty$ as $s \rightarrow 0$, some $s < u$ has $k_q(s) = \kappa$, where the value is $A/s - \kappa > A/u - \kappa$. So only u with $k_q(u) \geq \kappa$ matter, and there the value is $v_q(u)$. A point mass y_0 satisfying the constraints has $y_0 = E(Y) \leq -\kappa \leq 0 < L_q(\kappa)$, where positivity holds because $v_q(u) \sim u/2$ and $k_q(u) \sim A/u$ as $u \rightarrow 0$. Since $m_u(b) \geq \Phi(-b)$, $b_u(q) \geq -\Phi^{-1}(q)$, so k_q is bounded on $[u_0, \infty)$ for every $u_0 > 0$ and $k_q(u) \geq \kappa$ forces $u \rightarrow 0$ as $\kappa \rightarrow \infty$; moreover $ub_u(q) \rightarrow \log(1/q)$, so $k_q(u) = \{A + o(1)\}/u$, and $v_q(u) = u/2 + o(u)$ as $u \rightarrow 0$ (proof of Lemma 1). These expansions give $\kappa L_q(\kappa) \rightarrow A/2$: for large κ only small u are feasible, $u \leq \{A + o(1)\}/\kappa$, and $v_q(u) \leq \{1 + o(1)\}u/2$ with equality at the boundary. Monotonicity is clear, and $L_q(\kappa) = c_q$ while a maximizer of v_q is feasible. $\square$

Proposition S1. Let X have density proportional to $e^{\beta x}$ on $[-\ell, 0]$ with $\beta > 0$ and $0 < \ell \leq \infty$, let $W = X - E(X)$ and let $r_0 = Q_q(|W|)$. If $\pm r_0$ lie in the interior of the support of W , then, after scaling W so that the noisy q -quantile of $|W + x^{1/2}Z|$ equals 1,

$$Q_q(|W|) = 1 + \frac{1}{2}\beta r_0 \tanh(\beta r_0)x + o(x) \quad (x \rightarrow 0).$$

For every $q \in (0, 1)$ the condition holds for small $\beta\ell$, so $R_{q,q}^{(0)}(x) - 1$ is of exact order x .

Numerically the best coefficient in this family is 0.3466 at $q = 0.8$, for $W = 1 - E_1$, and 0.0747 at $q = 0.9$.

Proof of Proposition S1. Near $\pm r_0$ the density f of W is smooth with $f' = \beta f$, and the ends of the support are at positive distance. The jumps of the density at the ends of the support contribute $O\{v^{1/2}\phi(d_0/v^{1/2})\}$ near $\pm r_0$, where $d_0 > 0$ is the distance from $\pm r_0$ to the ends, which is $o(v)$. Hence $H_v(r) = \text{pr}(|W + v^{1/2}Z| \leq r) = H_0(r) + v\{f'(r) - f'(-r)\}/2 + o(v)$ uniformly near r_0 , and the noisy quantile is $t(v) = r_0 - v\beta\{f(r_0) - f(-r_0)\}/[2\{f(r_0) + f(-r_0)\}] + o(v) = r_0 - (\beta v/2) \tanh(\beta r_0) + o(v)$. Scaling by $t(v)$ and writing $x = v/t(v)^2$, a continuous and increasing function of v near 0, gives the expansion. As $\beta\ell \rightarrow 0$, W/ℓ tends to the uniform law on $[-1/2, 1/2]$, whose q -quantile of $|W|/\ell$ is $q/2 < 1/2$. $\square$

Proof of Theorem 2. Let $\mu = E(W)$. If W is a point mass, $Q_q(|W|) = |\mu| \leq B$. Otherwise, for $s > 0$ the law of $W + s^{1/2}Z$ is in $\mathcal{B}$ (Saumard, 2019), with mean μ , distribution function F_s and a positive smooth density f_s . For $r > \mu$ let $h = f_s(r)/F_s(r)$. The tangent of $\log F_s$ at r gives $F_s(y) \leq \min\{F_s(r)e^{h(y-r)}, 1\}$, the distribution function of an exponential tail with mean $r - \{1 + \log F_s(r)\}/h$ that lies stochastically below $W + s^{1/2}Z$, so $\mu \geq r - \{1 + \log F_s(r)\}/h$ and $h(r - \mu) \leq 1$. Laws in $\mathcal{B}$ satisfy $f'_s \leq f_s^2/F_s$ (Dümbgen et al., 2017), so $(\log f_s)'(r) \leq h \leq 1/(r - \mu)$; applied to $-W$ this gives $(\log f_s)'(-r) \geq -1/(r + \mu)$. Hence, for $r > B$,

$$f'_s(r) \leq \frac{f_s(r)}{r - B}, \quad -f'_s(-r) \leq \frac{f_s(-r)}{r - B}.$$

Let $H_s(r) = \text{pr}(|W + s^{1/2}Z| \leq r)$. By the heat equation $\partial_s f_s = f''_s/2$,

$$\partial_s H_s(r) = \frac{1}{2}\{f'_s(r) - f'_s(-r)\} \leq \frac{f_s(r) + f_s(-r)}{2(r - B)} = \frac{\partial_r H_s(r)}{2(r - B)}.$$

The quantile $r(s) = Q_q(|W + s^{1/2}Z|)$ solves $H_s\{r(s)\} = q$ and is continuously differentiable on $s > 0$, with $r'(s) \geq -1/[2\{r(s) - B\}]$ where $r(s) > B$. Hence $\{r(s) - B\}_+^2$ has derivative at least -1 almost everywhere, and $\{r(\epsilon) - B\}_+^2 \leq \{r(\sigma^2) - B\}_+^2 + \sigma^2 - \epsilon$. As $\epsilon \rightarrow 0$, $r(\epsilon) \rightarrow Q_q(|W|)$, because a nondegenerate law in $\mathcal{B}$ has a positive density on the interior of its support (Dümbgen et al., 2017), so the distribution function of $|W|$ is continuous and strictly increasing where it lies strictly between 0 and 1. Since $r(\sigma^2) \leq t$, the first inequality follows, and the second from $(d^2 + \sigma^2)^{1/2} \leq d + \sigma^2/(2d)$. $\square$

Proof of Theorem 3, first part. Let $V = W + x^{1/2}Z$ with distribution function F . Since $\log F$ and $\log(1 - F)$ are concave and $F(V)$ is uniform, $E\{\log F(V)\} = -1$ and Jensen's inequality gives $F\{E(V)\} \geq e^{-1}$ and likewise $1 - F\{E(V)\} \geq e^{-1}$. If $E(W) \geq 1$ then $\text{pr}(|V| \leq 1) \leq F\{E(W)\} \leq 1 - e^{-1} < q$, and similarly if $E(W) \leq -1$; so $|E(W)| < 1$. Theorem 2 with $B = |E(W)|$ and $t = 1$ gives $\Delta \leq (d^2 + x)^{1/2} - d$ with $d = 1 - |E(W)|$, which is (2). $\square$

Proof of Theorem 3, the limit. For the lower bound in the limit, take u with $k_q(u) \geq \kappa$ and the law $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$ of the lower bound in Theorem 1. Its mean is $1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1}\} \leq 1 - \kappa x^{1/2}$, because b_u decreases in its argument, and it is positive for small x ; hence $\liminf_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \geq v_q(u)$ for every such u .

For the upper bound, suppose that $x_n \rightarrow 0$, $h_n = x_n^{1/2}$, and that W_n is feasible for $R_{q,q}^{(b_{x_n})}(x_n)$ with $Q_q(|W_n|) - 1 \geq a_0 h_n$, where $L_q(\kappa) < a_0 < 1$. By (2), $\kappa h_n \leq 1 - |E(W_n)| \leq \bar{\kappa} h_n$ with $\bar{\kappa} = (1 - a_0^2)/(2a_0)$. Replacing W_n by $-W_n$ if necessary, $E(W_n) \geq 0$; let $Y_n = (W_n - 1)/h_n$, so that $-\bar{\kappa} \leq E(Y_n) \leq -\kappa$. The law of $T_n = Y_n + Z$ is in $\mathcal{B}$, with a smooth positive density of maximum M_n and distribution function F_n , and $F_n(0) \geq \text{pr}(|W_n + h_n Z| \leq 1) \geq q$ while $F_n\{E(T_n)\} \leq 1 - e^{-1}$ by the first part. So $q - 1 + e^{-1} \leq F_n(0) - F_n\{E(T_n)\} \leq M_n \bar{\kappa}$. For a law in $\mathcal{B}$ with smooth density g , maximum M and median m , the hazard $g/(1 - F)$ is nondecreasing, so $g(y) \leq 2g(m)\{1 - F(y)\} \leq 2g(m)$ for $y \leq m$, and g/F is nonincreasing, which gives the same for $y \geq m$; so $g(m) \geq M/2$; the tangent bounds for $\log F$ and $\log(1 - F)$ at m give $\text{pr}(|T - m| > y) \leq e^{-2g(m)y}$, hence $\text{var}(T) \leq 1/\{2g(m)^2\} \leq 2/M^2$. Thus $\text{var}(Y_n) \leq 2\bar{\kappa}^2/(q - 1 + e^{-1})^2$, the Y_n are tight and uniformly integrable, and along a subsequence $Y_n \rightarrow Y$ in distribution with $Y \in \mathcal{B}$,

possibly a point mass, since $\mathcal{B}$ is closed under weak convergence (Saumard, 2019), $E(Y) \leq -\kappa$ and $\text{pr}(Y + Z \leq 0) = E\{\Phi(-Y)\} \geq q$. On $\{W_n \geq 0\}$, whose probability tends to one, $(|W_n| - 1)/h_n = Y_n$, so $a_0 \leq \{Q_q(|W_n|) - 1\}/h_n \rightarrow Q_q(Y) \leq L_q(\kappa)$ by Lemma 2, where the quantile converges because a law in $\mathcal{B}$ has a unique q -quantile or is a point mass. This contradicts $a_0 > L_q(\kappa)$, so $\limsup_{x \rightarrow 0} x^{-1/2} \{R_{q,q}^{(b_x)}(x) - 1\} \leq a_0$ for every $a_0 \in (L_q(\kappa), 1)$; if $L_q(\kappa) \geq 1$ there is nothing to prove, because (2) gives $\Delta < x^{1/2}$. $\square$

Ranks for Proposition 3

At $q = 0.9$ and $\delta = 0.05$ no rank satisfies $p_k \geq 0.901$ and $k \geq Kp_k + 1$ for $K \leq 28$, since even $k = K$ gives $p_K = 0.05^{1/K} < 0.901$. For every K from 29, where the rank is $k = K$, to 3000 a rank exists, and it is at most three ranks above the usual rank with $p_k \geq 0.9$ (computed for every K); for instance 46 against 45 at $K = 46$ and 2731 against 2728 at $K = 3000$. The gap grows with K , to 10 at $K = 10^4$ (9060 against 9050), because the slack 10^{-3} becomes large relative to the spacing of the levels p_k . The function `certified_rank` returns this rank.

Proof of Proposition 2. $\bar{F}(T) < p_k$ exactly when $N = \#\{i : |V_i| < \bar{F}^{-1}(p_k)\} \geq k$. N is a sum of independent indicators whose success probabilities average p_k , and $k - 1 \geq Kp_k$. By Hoeffding (1956), $\text{pr}(N \leq k - 1)$ is smallest when all success probabilities are equal. Hence $\text{pr}(N \geq k) \leq \text{pr}\{\text{Bin}(K, p_k) \geq k\} = \text{pr}\{\text{Beta}(k, K + 1 - k) \leq p_k\} = \delta$. $\square$

The case $p > q$

The constant of § 4.1 is

$$C_{p,q} = \sup_{\alpha_L + \alpha_R = 1-p} \inf_{\beta_L + \beta_R = 1-q} \max(c_{1-\alpha_L, 1-\beta_L}, c_{1-\alpha_R, 1-\beta_R}).$$

Here $c_{1,q'} = -\infty$, since no law has noisy coverage one at a single end. Numerically the worst split puts the whole budget on one end, so that $C_{p,q} = c_{p,q}$; this observation is not used. Ball arithmetic gives $-0.0572 \leq C_{0.9036, 0.9} \leq -0.057$, where 0.9036 is the order-statistic level for $K = 110$, $k = 105$ at $\delta = 0.04$, used in § S5.

In the proof of the upper bound of Theorem 1, replace $1 - q$ by $1 - p$ for the noisy budget, choose any split $\beta_L + \beta_R = 1 - q$ of the latent budget, and use $F_Y^{-1}(1 - \beta_R) \leq c_{1-\alpha_R, 1-\beta_R}$ at each end. Since $c_{p',q'}$ decreases in p' , the worst case has $\alpha_L + \alpha_R = 1 - p$, which gives $R_{p,q}(x) \leq 1 + C_{p,q}x^{1/2}$ for every x , with $C_{p,q}$ nonincreasing in p . The proof of the lower bound applies with $b_u(q)$ replaced by $b_u(p)$ and $p' = p + \eta_x$, giving $R_{p,q}(x) \geq 1 + c_{p,q}x^{1/2} - o(x^{1/2})$ for every $p \geq q$. This proves Corollary 1, whose numerical intervals are certified in § S4.

Bi-log-concave laws

Corollary S1. *Let $\mathcal{B}$ be the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave (Dümbgen et al., 2017); $\mathcal{L} \subset \mathcal{B}$. Define $R_{p,q}^{\mathcal{B}}$ by (1) with $\mathcal{B}$ in place of $\mathcal{L}$. Then Lemma 1 holds over $\mathcal{B}$ with the same $c_{p,q}$, and for $p \geq q$ and every $x > 0$, $R_{p,q}^{\mathcal{B}}(x) \leq 1 + C_{p,q}x^{1/2}$, while $R_{q,q}^{\mathcal{B}}(x) = 1 + c_{q,q}x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$. In particular Corollary 1 holds over $\mathcal{B}$.*

Proof. The proof of Lemma 1 uses only the concavity of $\log F_Y$ at the q -quantile, so its upper bound holds for every Y with $\log F_Y$ concave, and exponential tails are in $\mathcal{L}$, which gives equality. In the proof of Theorem 1, $Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave, and $(-W - 1)/x^{1/2}$ has a concave log distribution function because $\log(1 - F_W)$ is concave and F_W is continuous. The lower bound uses laws in $\mathcal{L} \subset \mathcal{B}$. $\square$

The class $\mathcal{B}$ contains multimodal laws. Proposition 1 and the box-bound upper bounds of § S4 use log-concavity of the density; the ball-arithmetic constants do not (Corollary S1). Theorems 2 and 3, Lemma 2, Proposition 3 and Corollary S2 hold over $\mathcal{B}$: their proofs use the monotone hazards of $\mathcal{B}$, the bound $f' \leq f^2/F$ of Dümbgen et al. (2017), and the preservation of $\mathcal{B}$ under convolution with a log-concave law and under weak limits (Saumard, 2019). Both are elementary. If $\log F_X$ is concave and g is a log-concave density, $(t, y) \mapsto F_X(t - y)g(y)$ is log-concave on $\mathbb{R}^2$, so $F_{X+Y}(t) = \int F_X(t - y)g(y) dy$ is log-concave by the theorem of Prékopa (1973), and likewise $1 - F_{X+Y}$. For weak limits, concavity of $\log F_n$ and $\log(1 - F_n)$ passes to the limit at continuity points, and a limit that is not a point mass has no atom, because a concave function is continuous on the interior of its domain.

Symmetry alone does not make noisy calibration conservative in $\mathcal{B}$. The density $f(w) = \{1 + 0.05 \cos(2\pi w)\}/2$ on $(-1, 1)$ is in $\mathcal{B}$, because $\sup |f'| = 0.05\pi < \inf f^2 = 0.2256$ (Dümbgen et al., 2017, Theorem 1), but it is not unimodal. At $q = 0.7$, $r_0 = Q_q(|W|) \approx 0.7077$ and $f'(r_0) > 0$; since f' is odd, $\partial_s \text{pr}(|W + s^{1/2}Z| \leq r_0) = f'(r_0) > 0$ at $s = 0$, so noise raises the coverage and the noisy quantile falls below r_0 , to 0.707676 at $s = 10^{-5}$, against $r_0 = 0.707678$. Anderson’s comparison needs unimodality. The bimodal law of Table 1, the mixture $\pi N(m_1, s^2) + (1 - \pi)N(m_2, s^2)$ with mean zero, variance one and $(m_2 - m_1)/s = 2.5$, is in $\mathcal{B}$ by the following lemma. With $\pi = 0.4$ the mixture stays in $\mathcal{B}$ up to a separation of about 2.59, while bimodality starts at about 2.443 (numerical). The parameters are $\pi = 0.4$, $m_1 = -0.94868$, $m_2 = 0.63246$ and $s = 0.63246$.

Lemma S1. *Let f be the density of $\pi N(m_1, s^2) + (1 - \pi)N(m_2, s^2)$ with $m_1 < m_2$ and $r = (m_2 - m_1)/s > 2$, let $P(w)$ be the posterior probability of the first component, and let $a < b$ solve $P(a) = \frac{1}{2} + h$ and $P(b) = \frac{1}{2} - h$, $h = (1/4 - 1/r^2)^{1/2}$. The mixture is in $\mathcal{B}$ if $f'F < f^2$ on $[a, m_2]$ and $-f'(1 - F) < f^2$ on $[m_1, b]$.*

Proof. Since $(\log f)'' = -s^{-2} + P(1 - P)r^2s^{-2}$ and P decreases, f is log-concave on $(-\infty, a]$ and on $[b, \infty)$. On $(-\infty, a]$, $F(t) = \int 1(y \leq t)f(y)1(y \leq a) dy$ is the marginal of a log-concave function of (t, y) , so $\log F$ is concave there (Prékopa, 1973); likewise $\log(1 - F)$ on $[b, \infty)$. Where $f' \leq 0$, in particular on $[m_2, \infty)$, $f'F \leq f^2$ is trivial, and where $f' \geq 0$, in particular on $(-\infty, m_1]$, so is $-f'(1 - F) \leq f^2$. With the two hypotheses, $(\log F)'' \leq 0$ and $(\log(1 - F))'' \leq 0$ on the whole line. $\square$

For the law of Table 1 the two hypotheses are proved in Arb ball arithmetic on 128 boxes, with the ends a and b rounded outward (`uai.interval.bimodal_blc_certificate`, test suite); the largest values of $(f'F - f^2)/f^2$ on $[a, m_2]$ and of $\{-f'(1 - F) - f^2\}/f^2$ on $[m_1, b]$ are -0.141 and -0.864 . The bound of Theorem 2 with $E(W) = 0$ holds for this law and its version with the cosine phase-shifted by $\pi/2$ and recentred to mean zero, at $q = 0.3, 0.7, 0.9$ and $s = 10^{-3}, 0.1, 1$ (test suite).

S2 Non-Gaussian noise

Proposition 5 at both ends

For a noise law ϵ write $c_{p,q}(\epsilon)$ for the constant of Lemma 1 with Z replaced by ϵ ; the proof of Lemma 1 and of the upper bound of Theorem 1 used only that the exponential tail lies stochastically below Y , so they hold for any noise independent of W , with $c_{p,q}(\epsilon)$ at the right end and $c_{p,q}(-\epsilon)$ at the left. By Proposition 5, $c_{p',q'}(\epsilon) \leq 0$ for every ϵ in a class $\mathcal{E}$ closed under scaling and negation whenever $\Psi_{\mathcal{E}}(q') < p'$. Let $\lambda_{\mathcal{E}}(p') = \sup\{q' : \Psi_{\mathcal{E}}(q') < p'\}$, nondecreasing because $\Psi_{\mathcal{E}}$ increases in q , and $\phi(a) = 1 - \lambda_{\mathcal{E}}(1 - a)$, the latent budget at one end when the noisy budget there is a ; certificates use computed levels below $\lambda_{\mathcal{E}}$. If $\phi(a_L) + \phi(a_R) \leq 1 - q$ whenever $a_L + a_R \leq 1 - p$, then, as in the proof of Corollary 1, every $W \in \mathcal{B}$ with $\text{pr}(|W + e| \leq 1) \geq p$ has $\text{pr}(|W| \leq 1) \geq q$ for every $e = \sigma\epsilon$, $\epsilon \in \mathcal{E}$, $\sigma > 0$. In the proof of Proposition 3 the noise laws of the areas enter one at a time, so they may differ between areas. Since ϕ is nondecreasing it suffices to check finitely many splits, as in § S4.

Symmetric unimodal noise

By the theorem of Khintchine, a law symmetric and unimodal about 0 is that of RU , with U uniform on $[-1, 1]$ and $R \geq 0$ independent of U . Hence $E\{\min(1, qe^{-RU})\} = E\{g(R)\} \leq \sup_r g(r)$, with $g(r) = E\{\min(1, qe^{-rU})\}$, and uniform laws attain the supremum. With $c = \log q$,

$$g(r) = q \frac{\sinh r}{r} \quad (r \leq -c), \quad g(r) = \frac{c + r + 1 - qe^{-r}}{2r} \quad (r \geq -c).$$

The first branch increases. On the second, $g'(r)$ has the sign of $D(r) = qe^{-r}(1+r) - (1+c)$, which decreases, with $D(-c) = f(c) = e^{2c}(1-c) - 1 - c > 0$ because $f(0) = f'(0) = 0$ and $f''(c) = -4c e^{2c} > 0$ for $c < 0$, and $D(r) \rightarrow -(1+c) < 0$. So g is maximal at the root r of D , where $g(r) = \{1 + qe^{-r}\}/2$. At $q = 0.8, 0.9$ and 0.95 , $\Psi(q) = 0.80772, 0.90179$ and 0.95043 . The split certificates of § S4, with ϕ bounded through enclosures of Ψ , give $p_E \leq 0.8078, 0.9018$ and 0.9505 , and enclosures give $\Psi(0.8) > 0.8077$, $\Psi(0.9) > 0.9017$ and $\Psi(0.95) > 0.9504$, so these levels cannot be lowered to the next multiple of 10^{-4} . Noisy level 0.9 certifies latent level 0.8981, while $\Psi(0.8982) > 0.9$. Heavy tails are allowed: the t_3 and Cauchy laws are symmetric and unimodal, and for them the supremum over the scale is 0.90013 and 0.9 at $q = 0.9$ (numerical).

Mean-zero log-concave noise

Reduction. Let $h(e) = \min(1, qe^{-e})$, which is continuous, nonincreasing and 1-Lipschitz. If ϵ is log-concave with mean zero, conditioning on $|\epsilon| \leq M$ and recentring changes $E\{h(\epsilon)\}$ by at most $2\text{pr}(|\epsilon| > M) + |E(\epsilon | |\epsilon| \leq M)|$, which tends to zero; so it suffices to bound $E\{h(\epsilon)\}$ over log-concave laws on compact intervals with $E(\epsilon) \geq 0$, as h is nonincreasing. By the localization theorem (Fradelizi & Guédon, 2004) the supremum of this linear functional is attained at a point mass δ_x with $x \geq 0$, where $h(x) \leq q$, or at a law with density proportional to $e^{\beta e}$ on a segment with mean zero. Write the latter as $\epsilon = L(V - \mu_k)$, V on $[0, 1]$ with density proportional to e^{kv} , $k = \beta L$, $\mu_k = E(V)$; then

$$E\{h(\epsilon)\} = \frac{1}{Z_k} \left\{ \int_0^{v_c} e^{kv} dv + qe^{L\mu_k} \int_{v_c}^1 e^{(k-L)v} dv \right\}, \quad v_c = \min\{1, \max(0, \mu_k + c/L)\},$$

with $Z_k = \int_0^1 e^{kv} dv$. For $|k| \geq 2$ we use the scale $\sigma = L/|k|$ and the truncated exponential form $\epsilon = \pm\sigma(\tau - m)$, τ standard exponential truncated to $[0, |k|]$, which includes $|k| = \infty$, the centred exponential laws $\pm\sigma(E - 1)$.

Certificate. A branch and bound in Arb ball arithmetic, `uai.noise.classes.lc.certificate.parallel`, bounds $E\{h(\epsilon)\}$ from above on boxes of (k, L) for $|k| \leq 2$ and of $(1/|k|, \sigma)$ for $|k| \geq 2$, splitting until each box is below the target. On a box the bound is $\Phi(u) = \text{pr}(\epsilon \leq u) + qE\{e^{-\epsilon}; \epsilon > u\}$ at one cut u , the kink c of the box centre, since $h(e) \leq 1(e \leq u) + qe^{-e}1(e > u)$ for every u ; this removes the dependence of the kink on the parameters, and $\Phi'(c) = 0$ makes the bound tight to second order; Z_k and $\int_0^1 v e^{kv} dv$ are evaluated by their series with a remainder bound near $k = 0$. The ends of the scale parameter are closed analytically: for small L , Hoeffding's lemma gives $E\{h(\epsilon)\} \leq q E(e^{-\epsilon}) \leq qe^{L^2/8}$; for large L , $E\{h(\epsilon)\} \leq \text{pr}(\epsilon \leq 4) + qe^{-4}$, where $\text{pr}(V \leq \mu_k) \leq 1 - e^{-1}$ as in the proof of Theorem 3 and the density of V is at most $2/(1 - e^{-2})$; for small and large σ the analogous bounds use $E(e^{-\sigma\tau}) \leq 1$, $E(e^{\sigma\tau} | \tau \leq |k|) \leq (1 - \sigma)^{-1}$ and $m \geq 1 - 2/(e^2 - 1)$. At $q = 0.9$ the certificate gives $\Psi_E(0.9) < 0.9052$, and the centred exponential with $\sigma = 0.11777$ gives $\Psi_E(0.9) > 0.90517$; it gives $\Psi_E(0.8942) > 0.9$ at $\sigma = 0.1259$. For the two-sided levels, a partition of the noisy budget $a_L \in [0, (1 - p)/2]$ is proposed from the centred exponential values, so that each interval uses at most half the room, and every node a is certified by the branch and bound at a level q_a below $\lambda_E(1 - a)$; this certifies the noisy levels 0.9068 and 0.906 for latent level 0.9, with 29 and 41 intervals, and noisy level 0.9 for latent level 0.8935, with 45 intervals (`split_certificate_lc`). Numerically the worst split

puts the whole budget at one end, as for Gaussian and symmetric unimodal noise, so the two-sided level is $\Psi_{\mathcal{E}}(0.9) = 0.905176$. Numerically, along truncated exponential laws $\sigma(\tau - m)$ with a long right tail the supremum over the scale increases with the truncation length, to the centred exponential; for the reflected laws it decreases.

Mean-zero unimodal noise

Let ϵ have density $(1 - \rho)/a$ on $[-a, 0]$ and ρ/b on $(0, b]$ with $b = (1 - \rho)a/\rho$ and $\rho \leq 1/2$; it is unimodal with mean zero. Then $E\{h(\epsilon)\} \geq (1 - \rho)(1 + c/a)$, which tends to one as $\rho \rightarrow 0$ and $a \rightarrow \infty$, so $\Psi_{\mathcal{E}}(q) = 1$ and no noisy level below one transfers to any latent level.

Gaussian noise

For Gaussian noise $\Psi_{\mathcal{E}}(q) = \sup_{\sigma} E\{\min(1, qe^{-\sigma Z})\}$, and Proposition 5 recovers the sign of the constants of Lemma 1: $c_{p,q} \leq 0$ if $\Psi_{\mathcal{E}}(q) < p$ and $c_{p,q} > 0$ if $\Psi_{\mathcal{E}}(q) > p$. At $q = 0.9$ the supremum is 0.900803, at $\sigma = 0.0503$, in the slack interval $(0.90079, 0.901]$ of Corollary 1.

S3 The shape of the sharp radius

Figure S1 shows $R_{q,q}(x) - 1$ computed by grid maximization over the extremal family; grid values bound $R_{q,q}$ from below. The levels at which the grid value crosses 1 are $x^*(q) \approx 0.034, 0.017, 0.0062$ and 0.0012 for $q = 0.8, 0.85, 0.9$ and 0.95 , and the largest widening found is 0.37%, 0.17%, 0.063% and 0.012%. At $q = 0.9$ the numerical upper bounds of § S4 give $R_{q,q}(x) \leq 1$ for $x \geq 0.013$, so Theorem 1 bounds the widening by $c_{0.9} 0.013^{1/2} < 0.22\%$ at every x ; this rests on the double-precision bounds, whereas the bound 1.16% of § 3 rests only on Theorem 1 and the certified $c_{0.9}$. Below $x^*(q)$ the noisy threshold must be widened; above it, grid values below 1 do not by themselves show that shrinking is safe, which the numerical upper bounds of § S4 do. The exact grid values lie below $1 + c_{0.9}x^{1/2}$ at $x = 10^{-6}, \dots, 10^{-2}$, as Theorem 1 requires.

Centred laws and the transition

The following values are computed in double precision by `experiments/e32_centering.py` and are not certified. For $q = 0.8, 0.9$ and 0.95 : $\kappa_q^* = 6.8118, 20.1823$ and 50.0741 , attained at $u^* = 0.11328, 0.044286$ and 0.018943 . The two upper bounds in Fig. S2 cross at $\kappa = 10.829, 26.221$ and 59.466 , where L_q/U_q is smallest, 0.770, 0.879 and 0.927; as $\kappa \rightarrow \infty$ the ratio tends to $1 - \log(1/q) = 0.777, 0.895$ and 0.949 . At $q = 0.9$, $L_q(25) = 0.017393$ and $L_q(50) = 0.0089448$. For the mean-zero laws of Proposition S1, the coefficient with $\beta = 0.2$ and $\ell = 2$ at $q = 0.9$ is 0.0156540; the expansion is accurate to 6×10^{-5} at $x = 1.3 \times 10^{-4}$ but fails at $x \approx 1.1 \times 10^{-3}$, where the noise reaches the end of the support and the threshold may already be shrunk. The bound of Theorem 2, including its form with $|E(W)| \leq B$, was checked on 642 laws from the family of Proposition S1, with no violation.

The location of the mean in examples

Figure S3 shows the radius correction for laws scaled so that the noisy coverage of $[-1, 1]$ is exactly 0.9, computed from closed forms by `experiments/e35_mean_location.py`. For the boundary-layer law $W = 1 + x^{1/2}\{b - E_u\}$ at the maximizer u^* of v_q , whose mean lies $20.18 x^{1/2}$ from the boundary, the correction divided by $x^{1/2}$ is 0.019062 for $x \leq 3 \times 10^{-5}$, equal to $c_{0.9}$. For the same family at the rate with $k_q(u) = 50$ it is $0.008945 = L_{0.9}(50)$. For the fixed mean-zero law of Proposition S1 with $\beta\ell = 0.5$ on a unit segment,

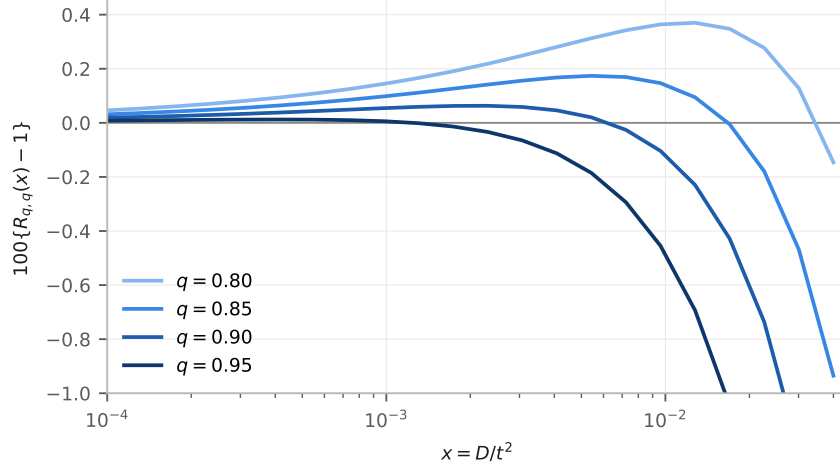


Figure S1: Sharp latent radius relative to the noisy threshold, $100\{R_{q,q}(x) - 1\}$, for four coverage levels, computed by grid maximization over the extremal family, which bounds $R_{q,q}$ from below.

Alt text: Line plot of the percentage radius correction against the noise ratio for four coverage levels; each curve is positive for small noise ratios, peaks below half a per cent and turns negative beyond a level-dependent crossing point.

the correction divided by x is 0.024005 up to $x = 5 \times 10^{-5}$, and it turns negative once the noise reaches the end of the support. Larger centred coefficients sit closer to that end: at $\beta\ell = 0.9$ the coefficient is 0.070 but the quantile lies 0.0011 from the end, and the correction is positive only for x below about 10^{-7} . A Gaussian law with mean $1.29x^{1/2}$ from the boundary after scaling, $W \sim N(1 - 2x^{1/2}, 0.01x)$ before scaling, needs no correction at all: its radius is $1.16x^{1/2}$ below the noisy threshold. A mean near the boundary is necessary for a correction of order $x^{1/2}$, not sufficient.

The limit with κ fixed

The limit in Theorem 3 is taken with κ fixed. Applying the small- u construction to a centred law, that is, $\kappa = x^{-1/2}$, gives in the limit a left exponential tail of rate $1 - \log(1/q)$ whose noise-free central coverage is $q\{1 - e^{-2+2\log(1/q)}\}$, 0.750 at $q = 0.9$, far below the constraint. So $\{1 - \log(1/q)\}/2$ is not a lower bound for the centred coefficient.

S4 Certification

Box bounds for the radius

By the symmetry $W \mapsto -W$ of the kernel and of $|W|$ it suffices to treat $\beta \geq 0$. Write a member of $\mathcal{F}$ with $\beta \geq 0$ as $W = b - Y$, where Y has density proportional to $e^{-\beta y}$ on $[0, \ell]$. The ends of the family are included: $\ell = 0$ or $\beta = \infty$ gives a point mass and $\ell = \infty$ an exponential tail. For slopes $\beta' > \beta$ the likelihood ratio $e^{-(\beta' - \beta)y}$ is decreasing, so $Y_{\beta'}$ lies below Y_{β} in likelihood-ratio order, and for lengths $\ell < \ell'$ truncation gives $F_{\ell} \geq F_{\ell'}$. Thus W is stochastically increasing in β and b and decreasing in ℓ . On a box of parameters let W_{lo} and W_{hi} be the stochastically smallest and largest corners. Since $\text{pr}(W + e \leq \pm 1)$ and $\text{pr}(W \leq y)$ are expectations of decreasing functions of W , every law in the box satisfies

$$\text{pr}(|W + e| \leq 1) \leq \text{pr}(W_{\text{lo}} + e \leq 1) - \text{pr}(W_{\text{hi}} + e \leq -1), \quad \text{pr}(|W| \leq s) \geq \text{pr}(W_{\text{hi}} \leq s) - \text{pr}(W_{\text{lo}} \leq -s).$$

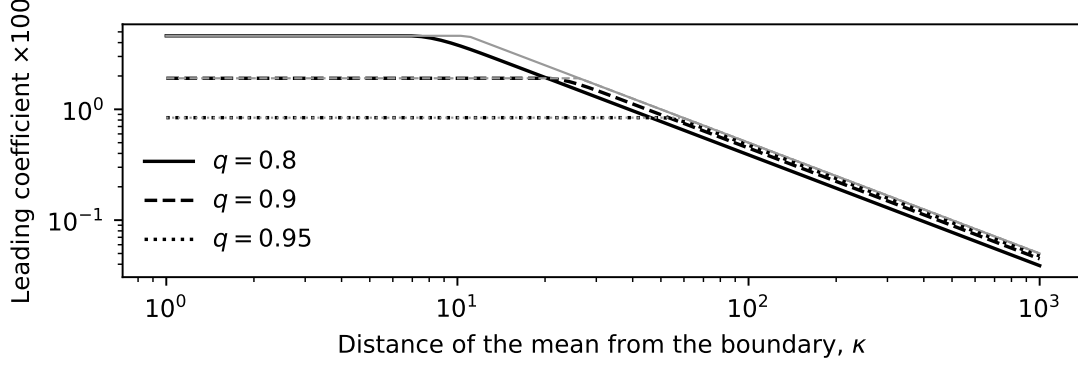


Figure S2: Leading coefficient $L_q(\kappa)$ of the worst-case radius correction, times 100, when the mean lies at distance at least $\kappa x^{1/2}$ from the boundary (Theorem 3), for $q = 0.8$ (solid), 0.9 (dashes) and 0.95 (dots), with the upper bound $\min\{c_q, (\kappa^2 + 1)^{1/2} - \kappa\}$ from Theorems 1 and 2 (grey). Both axes are logarithmic.

Alt text: Log-log plot of the leading coefficient against the scaled distance of the mean from the boundary for three levels; each curve is flat up to about 7, 20 and 50, then decays like the reciprocal of the distance, just below a grey upper bound.

The density bound $\Pr(|W + e| \leq 1) \leq 2 \sup f_W = 2\beta/(1 - e^{-\beta\ell})$ is also available. A box whose first bound is below p , or whose second bound is at least q , contains no law that violates $R_{p,q}(x) \leq s$, and a branch and bound over boxes certifies $R_{p,q}(x) \leq s$. The location is confined to $[-1 + x^{1/2}\Phi^{-1}(p), b_{\max}]$: the lower end holds because $W \leq b$; for the upper end take $k = 7x^{1/2}$; the density of W increases towards b , so $\Pr(|W| \leq 1 + k) \leq r/(1 + r)$ with $r = (2 + 2k)/(b - 1 - k)$, and $\Pr(|e| > k) = 2\Phi(-7)$. The parameters $\beta, \ell \in [0, \infty]$ are compactified.

Proposition S2. For all $x, h \geq 0$, $R_{p,q}(x + h) \leq R_{p,q}(x) + c_q h^{1/2}$.

Proof. If W is feasible at noise level $x + h$, then $W + h^{1/2}Z'$, with $Z' \sim N(0, 1)$ independent of (W, Z) , is log-concave and feasible at level x . So $s = Q_q(|W + h^{1/2}Z'|) \leq R_{p,q}(x)$, and Theorem 1 at scale s gives $Q_q(|W|) \leq s + c_q h^{1/2}$. $\square$

We computed numerical upper bounds of $R_{p,0.9}$ for $p = 0.9, 0.9036$ and 0.9068 on $0.002 \leq x \leq 0.366$ with step 5×10^{-4} , using Proposition S2 between grid points. Here 0.9068 and 0.9036 are the order-statistic levels p_k of $K = 110$, $k = 105$ at $\delta = 0.05$ and $\delta = 0.04$; the latter is used in § S5. The closed forms agree with 30-digit quadrature to within 10^{-11} at random points with $10^{-4} \leq x \leq 1.5$, $10^{-4} \leq \ell \leq 20$ or $\ell = \infty$ and $10^{-3} \leq \beta\ell \leq 10^3$, which cover the ranges in which boxes are evaluated after outward rounding, including the mixture kernels of Table 1, whose largest ratio D_i/T^2 is about 1.25 (test suite; the largest error found was 1.1×10^{-12} for $x \leq 0.37$ and 2.5×10^{-11} above), and boxes are cleared with a margin of 10^{-9} ; the bounds therefore rest on double-precision evaluation of the closed forms and are not interval arithmetic. For $0.05 \leq x \leq 0.3$ the bound exceeds a feasible value by at most a relative 3×10^{-4} for $p = 0.9$ and 10^{-3} for $p = 0.9068$. The branch and bound fails only just before the feasibility edges, near $x = 0.370$ for $p = 0.9$, on $0.3585 \leq x \leq 0.3615$ for $p = 0.9036$ and on $0.352 \leq x \leq 0.3545$ for $p = 0.9068$; there Proposition S2 from the last grid point gives $R_{0.9,0.9}(x) \leq 0.132$. Below $x = 0.01$ the lookup also uses Theorem 1 and its $p \geq q$ form, which are within 0.3% of a feasible value there (Fig. S4). At $q = 0.9$ the numerical bounds fall below 1 from $x = 0.013$ on, with the continuity margin $c_{0.9}(5 \times 10^{-4})^{1/2} \approx 4.3 \times 10^{-4}$ keeping them below 1 between grid points, and reach 0.973 at $x = 0.05$. A differential-evolution search over the three-parameter family, which evaluates the constraint by Gauss–Legendre quadrature, returned 1.027 at $x = 0.001$, above the bound 1.0006 of Theorem 1: the quadrature is inaccurate for steep slopes at small x . That search is

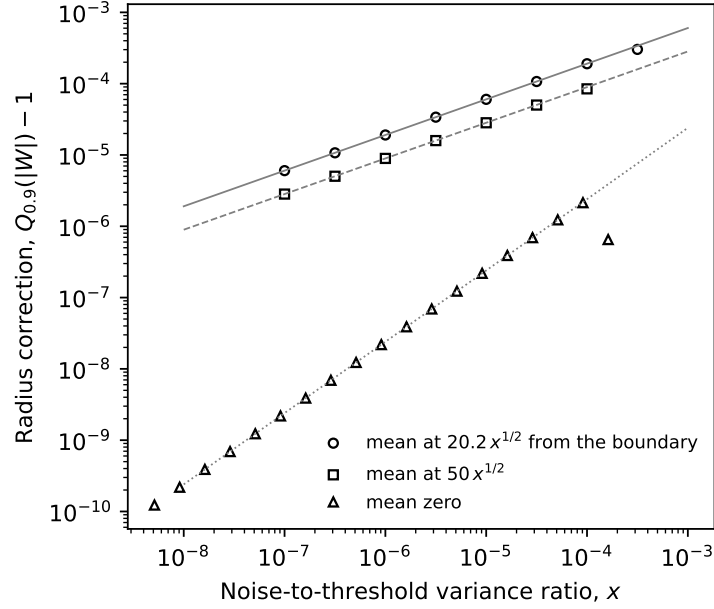


Figure S3: Radius correction $Q_{0.9}(|W|) - 1$ against x on logarithmic axes, for laws with noisy coverage exactly 0.9: mean at $20.2x^{1/2}$ from the boundary (circles), at $50x^{1/2}$ (squares) and mean zero (triangles), with $c_{0.9}x^{1/2}$ (solid), $L_{0.9}(50)x^{1/2}$ (dashes) and $0.024x$ (dots). Only positive corrections are shown.

Alt text: Log-log plot of the radius correction against the noise ratio for three laws; the two laws with mean near the boundary follow lines of slope one half and the mean-zero law follows a line of slope one.

therefore used only for $x \geq 0.006$, and the legacy table `results/r_table.json` built with it is not used in the paper.

Ball-arithmetic certificates for the constants

Every deciding inequality is evaluated in Arb ball arithmetic; floating point only proposes points, balls are converted to doubles with outward rounding, and sums of bounds are formed as balls.

1. $m_u(b)$ decreases in b , and in u because E_u is stochastically decreasing in its rate. Hence $b_u(p)$ is nonincreasing in u , and on $[u_0, u_1]$, $b_u(p) - u^{-1} \log(1/q) \leq b_{u_0}(p) - u_1^{-1} \log(1/q)$. A ball with $m_u(b) < p$ proves $b_u(p) < b$.
2. For $u \geq u_{\max}$ the value is at most $b_{u_{\max}}(p)$. For small u , let $b(u) = \{\log(1/p) + u^2/2 - \log(1 - u^3)\}/u$. Then $e^{-ub+u^2/2} = p(1 - u^3)$, and the Mills ratio gives $\Phi(-b) \leq pu^3$ whenever $\phi\{\log(1/p)/u\} \leq pu^2 \log(1/p)$. This holds on $(0, u_{\min}]$ once it holds at $u_{\min} \leq \log(1/p)/2^{1/2}$, because the left side divided by u^2 increases there. So $b_u(p) \leq b(u)$, and for $p \geq q$ the resulting bound increases in u .
3. By Lemma 1, $c_{p',q'} \leq \gamma$ if and only if $1 - q' \geq \varphi_\gamma(1 - p')$, where $\varphi_\gamma(a) = \max[0, \sup_u 1 - \exp\{-u(b_u(1 - a) - \gamma)\}]$. Hence $C_{p,q} \leq \gamma$ if $\varphi_\gamma(a_L) + \varphi_\gamma(a_R) \leq 1 - q$ whenever $a_L + a_R = 1 - p$. Since φ_γ is nondecreasing, it suffices to check this at the ends of an adaptive partition of a_L , with φ_γ bounded above by the same monotone branch and bound in u .

The certificates give $c_{0.8} \in [0.0460730, 0.0460731]$, $c_{0.9} \in [0.0190618, 0.0190619]$, $c_{0.95} \in [0.0084075, 0.0084077]$ and $c_{0.99} \in [0.0013950, 0.0013958]$, $-0.0571845 \leq c_{0.9036,0.9} \leq C_{0.9036,0.9} \leq -0.057$, $-0.1149813 \leq$

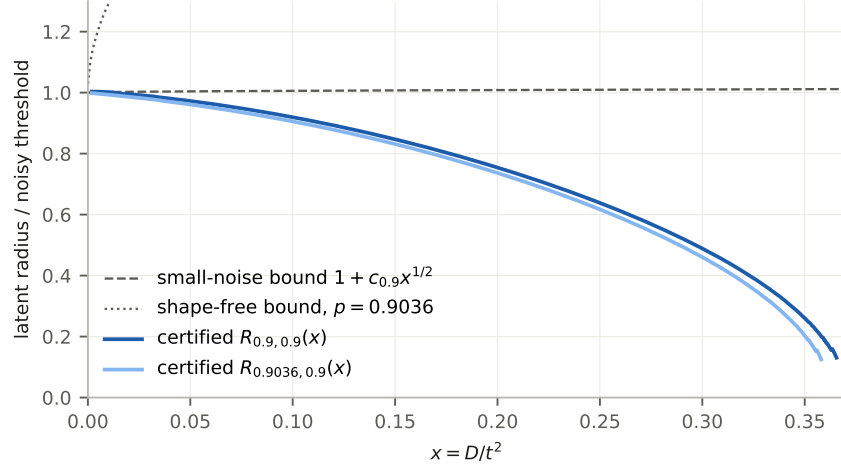


Figure S4: Numerical upper bound of the sharp radius $R_{p,0.9}(x)$ for $p = 0.9$ and for the order-statistic level $p = 0.9036$ ($\delta = 0.04$), with the small-noise bound of Theorem 1 and the shape-free bound $1 + x^{1/2}z_{1-(p-q)/2}$, which follows from $|W| \leq |V| + |e|$.

Alt text: Line plot of numerical upper bounds of the sharp radius against the noise ratio; the bounds lie above one only for small noise ratios and fall well below the shape-free bound.

$c_{0.9068,0.9} \leq C_{0.9068,0.9} \leq -0.114$, and $C_{p,q} \leq 0$ for $(q, p-q) = (0.8, 0.005)$, $(0.9, 0.001)$ and $(0.95, 0.0002)$. Lower bounds $c_{p,q} > 0$ at $p - q = 0.0043$, 0.00079 and 0.00015 come from single points u , evaluated in ball arithmetic, and $C_{p,q} \geq c_{p,q}$. Hence the smallest slack in Corollary 1 lies in $(4.3 \times 10^{-3}, 5.0 \times 10^{-3}]$, $(7.9 \times 10^{-4}, 1.0 \times 10^{-3}]$ and $(1.5 \times 10^{-4}, 2.0 \times 10^{-4}]$ at $q = 0.8, 0.9$ and 0.95 . For the coverage form of Corollary 1, the same certificates give $C_{0.8,0.795} \leq 0$, $C_{0.9,0.8991} \leq 0$ and $C_{0.95,0.9498} \leq 0$, and enclosures give $c_{0.8,0.7954} \geq 7.0 \times 10^{-4}$, $c_{0.9,0.8992} \geq 4.2 \times 10^{-4}$ and $c_{0.95,0.9499} \geq 3.0 \times 10^{-3}$, so by the lower bound of the case $p \geq q$, $R_{p,q'}(x) > 1$ for small x at these pairs (`experiments/e37_coverage_transfer.py`). Double-precision root finding puts the smallest slack at 4.4×10^{-3} , 8.0×10^{-4} and 1.6×10^{-4} . In coverage terms: as $x \rightarrow 0$ the boundary-layer laws $1 + x^{1/2}\{b_u(q) - E_u\}$, whose noisy coverage tends to exactly q , have latent coverage tending to $e^{-ub_u(q)}$, whose smallest values are 0.79532 , 0.89918 and 0.94984 at $q = 0.8, 0.9$ and 0.95 , attained at $u = 0.140$, 0.0508 and 0.0210 (double precision). Calibrating at the noisy level q thus costs at most a few thousandths of latent coverage in these examples, and the certified slack removes the loss.

S5 Heterogeneous and estimated noise variances

Heterogeneous known variances

In the heterogeneous rule, ε bounds, for each data set, the sup-norm error of the average kernel after compressing the D_i to 32 support points: the maximum over a grid of spacing $h = 10^{-3}$ plus $Mh^2/8$, with M a bound on the second derivative; it was about 2×10^{-4} . For each data set a numerical upper bound of R^{mix} is computed by the box bounds, which hold term by term for a mixture kernel, and the shape-free bound is used if the branch and bound fails.

Replacing the average kernel by the kernel of a single variance is not safe. With the mean variance plugged in, the radius is 1.4% too short at mean $x = 0.2$ with lognormal spread 1 (numerical). The smallest variance also fails: take $W = b - E_2$ with $b = 1.0435647$ and noise an equal mixture of $N(0, 10^{-8})$ and $N(0, 5 \times 10^{-4})$. The noisy coverage of $[-1, 1]$ exceeds 0.9, while the radius $1 + c_{0.9}10^{-4}$ given by the smaller

variance covers W with probability 0.89977; this was checked in 192-bit ball arithmetic.

A known lower bound on the variances

Corollary S2. *Let $G \in \mathcal{B}$ and $k \geq K p_k + 1$. If $D_i \geq D_{\min}$ for all i and $C_{p',q} < 0$ for some p' with $q \leq p' \leq p_k$, then $s = \max\{0, T + C_{p',q} D_{\min}^{1/2}\}$ satisfies $\text{pr}_{\mathcal{D}}\{\text{pr}(|W_{\text{new}}| \leq s \mid \mathcal{D}) \geq q\} \geq 1 - \delta$, whether or not the D_i are otherwise known.*

Proof. On the event of Proposition 2, some i has $\text{pr}(|W + e_i| \leq T) \geq p_k \geq p'$, and the bound of § 4.1 at $x = D_i/T^2$ gives $Q_q(|W|) \leq T + C_{p',q} D_i^{1/2} \leq T + C_{p',q} D_{\min}^{1/2}$. $\square$

At $q = 0.9$, $K = 110$ and $\delta = 0.05$, where $p_k = 0.906806$, ball arithmetic gives $C_{0.9068,0.9} \leq -0.114$, hence $s = T - 0.114 D_{\min}^{1/2}$, the shrunk threshold of Table 1.

Estimated variances

Suppose the D_i are unknown and estimates $\hat{D}_i$ are available that are independent of the scores, as for the sample means and variances of normal unit data. The order-statistic event of Proposition 2 involves the scores only. Let $\mathcal{E}_\eta$ be an event determined by the $\hat{D}_i$ with $\text{pr}(\mathcal{E}_\eta) \geq 1 - \eta$.

Lemma S2. *Let $G \in \mathcal{L}$. Suppose that on $\mathcal{E}_\eta$ a continuous kernel u_t , vanishing at $\pm\infty$, satisfies $u_t \geq \bar{g}_{D,t}$ pointwise for every $t > 0$. Then $s = T R_{p_k,q}[u_T]$, the radius (1) with kernel u_T , covers a new latent residual with probability at least q , conditionally on the data, with probability at least $1 - \delta - \eta$. If $\mathcal{D}_\eta$ is a confidence set with $\text{pr}(D \in \mathcal{D}_\eta) \geq 1 - \eta$, the same holds for $s = T \sup_{d \in \mathcal{D}_\eta} R_{p_k,q}^{\text{mix}}(d/T^2)$. With a compressed kernel, p_k is replaced by $p_k - \varepsilon$ as in § 4.2.*

Proof. On both events, $\int u_T dG \geq \int \bar{g}_{D,T} dG \geq p_k$, and Proposition 1 applies to u_T . The union bound gives $1 - \delta - \eta$, with no independence between the two events needed. $\square$

Lemma S3. *For $\lambda \geq 1$, $R_{p,q}^{\text{mix}}(\lambda x_1, \dots, \lambda x_K) \leq \lambda^{1/2} R_{p,q}^{\text{mix}}(x_1, \dots, x_K)$.*

Proof. If W is feasible at λx , then $U = \lambda^{-1/2} W$ satisfies the constraint at threshold $\lambda^{-1/2} \leq 1$, hence at threshold 1, so $Q_q(|W|) = \lambda^{1/2} Q_q(|U|) \leq \lambda^{1/2} R_{p,q}^{\text{mix}}(x)$. $\square$

A confidence box for all K variances needs level $1 - \eta/K$ per area and is useless with one or two degrees of freedom. Survey practice instead models variances through a few parameters, as generalized variance functions do. In the simplest such model $D_i = \sigma^2 a_i$ with known a_i , for instance $a_i = 1/n_i$, and the pooled estimate satisfies $\nu \hat{\sigma}^2 / \sigma^2 \sim \chi_\nu^2$ with $\nu = \sum_i (n_i - 1)$. The confidence set is the segment $\{\sigma^2 a : \ell \leq \sigma^2 \leq h\}$. Lemma S3 covers the supremum over it by finitely many points without assuming that R is monotone in σ^2 : for $\ell = s_0 < \dots < s_m = h$,

$$\sup_{\ell \leq \sigma^2 \leq h} R_{p_k,q}^{\text{mix}}(\sigma^2 a / T^2) \leq \max_j (s_{j+1}/s_j)^{1/2} U_j, \quad U_j \geq R_{p_k,q}^{\text{mix}}(s_j a / T^2).$$

Plugging in the lower end ℓ alone would not be justified, since R can increase with the noise level (Theorem 1).

In simulations, areas have n_i normal units, so $D_i = \sigma_i^2/n_i$ and the within-area variances are independent of the direct estimates; $K = 110$, mean $D = 0.577$, and 100 replications per setting and law (normal, Laplace, truncated Laplace). The rule uses $\delta = 0.04$ and $\eta = 0.01$, split as 0.009 below and 0.001 above the pooled interval for σ^2 , and needs 18 grid points on average. The radii below use converged grid values at those points; for the first 30 data sets of every setting and law (270 in all) the rule was rerun with numerical upper bounds U_j at every grid point, as Lemma S2 requires. The branch and bound succeeded for every data set, and the half-width grew by 0.14% on average and by at most 1.0%, at about 3.6 minutes per data set on one core.

1. With $n_i \in \{2, 3\}$, $\nu \approx 165$, the rule is 6.3–6.5% shorter than noisy conformal calibration and 2.9–3.0% longer than the known-variance rule at the same level, which is 9.0–9.1% shorter than noisy conformal calibration. With $n_i \in \{5, \dots, 10\}$, $\nu \approx 740$, the figures are 7.1–7.7% and 1.6–1.7%. Relative to the known-variance rule at $\delta = 0.05$, which needs no budget for the variances and meets the same final target, 62–63% and 73–74% of its gain survive. Reliability of the rule with grid values was at least 0.98; with numerical upper bounds, on the 30 data sets per setting, it was at least 0.967.
2. The supremum over the interval exceeded the value at its lower end by 0.2%.
3. When the within-area variances differ, with lognormal spread 0.5, the common-variance model is wrong and no guarantee applies; the rule was still reliable, at least 0.99, and 5.9–6.5% shorter than noisy conformal calibration.
4. Without pooling, with each area’s variance estimated from ν own degrees of freedom, a kernel envelope built from area-wise intervals and a binomial bound on the number of misses is 31–40% longer than noisy conformal calibration at $\nu = 2$, within 1.3% of it at $\nu = 50$, and 4.7–5.0% and 6.6–7.2% shorter at $\nu = 200$ and 1000 (30 replications per law; grid values). Whether a better area-wise construction exists is open.

S6 The shape-free comparison rule

Proposition S3. *Let $\bar{g}$ be a kernel that is symmetric, continuous, strictly decreasing in $|w|$ and tends to 0 as $|w| \rightarrow \infty$, with $m = \bar{g}(0)$, and suppose that $qm < p < m$. Over all laws G on the real line with $\int \bar{g} dG \geq p$, the supremum of $Q_q(|W|)$ is the radius r that solves $\bar{g}(r) = (p - qm)/(1 - q)$.*

Proof. Let $\tau = (p - qm)/(1 - q) \in (0, m)$. Since $m - \bar{g}(W) \geq 0$, Markov’s inequality gives $\text{pr}\{\bar{g}(W) < \tau\} \leq \{m - p\}/(m - \tau) = 1 - q$, and $\bar{g}(W) < \tau$ exactly when $|W| > r$, so $Q_q(|W|) \leq r$. Conversely, for small $\epsilon > 0$ put mass $q - \epsilon$ at 0 and mass $1 - q + \epsilon$ at the point $w_\epsilon > 0$ with $\bar{g}(w_\epsilon) = \{p - (q - \epsilon)m\}/(1 - q + \epsilon)$. This law meets the constraint with equality, has $Q_q(|W|) = w_\epsilon$, and $w_\epsilon \rightarrow r$ as $\epsilon \rightarrow 0$. $\square$

The average kernel of § 4.2 satisfies these conditions, so for a fixed rank the shape-free rule of Table 1 is the smallest radius that the single order-statistic constraint allows without a shape assumption. With 1 in place of $\bar{g}(0)$ it contains the bound of Guille-Escuret & Ndiaye (2024). It is not a bound on procedures that use several thresholds or tune the rank on independent data, as LatentCP (Zheng et al., 2026) can.

S7 Simulation details and an application

Table 1

Conditional coverage is computed from closed-form latent distribution functions. With 150 data sets the binomial standard error of a reliability is 0.018 at 0.95 and at most 0.029 at the lowest reliability observed, 0.847; for instance $145/150 = 0.967$ has the 95% binomial interval $[0.924, 0.989]$. The computed radius uses the order-statistic level $p_k = 0.9068$ of $\delta = 0.05$. The branch and bound returned a numerical upper bound of R^{mix} for every one of the 1050 data sets, at relative tolerance 10^{-3} in 671, 3×10^{-3} in 363 and 10^{-2} in 16, so the shape-free fallback was never used. With the mean variance in a single Gaussian kernel, which is not a valid plug-in, the numerical radius at level p_k is 2.9–3.2% shorter than at level $p = q$ (LDC_cert_pk against LDC_PAC in `results/hetldc_synth_summary.csv`), so most of the gain over a $p = q$ calibration comes from the order-statistic level. The shape-free rule was run at every fixed rank 106–110 and all results are stored; rank 107 is reported because it is best on average over the first four laws, which favours the

comparison rule. The Fay–Herriot interval falls short of the target under the Laplace law by about 1.3 standard errors, under the truncated exponential law, with reliability 0.947, by 0.2, and under the truncated Laplace law, with reliability 0.847, by about 5.8. The Fay–Herriot tolerance interval is our adaptation of the parametric bootstrap of Chatterjee et al. (2008) to an intercept-only model: the bootstrap $(1 - \delta)$ -quantile of the multiplier each refit needs to cover $N(\hat{\mu}, \hat{A})$ with probability 0.9. The latent residuals are drawn directly, with no fitted centre, and the mean noise variance 0.577 matches the ratio of design to model variance in the application below.

Reliability of the analytic rules with 2000 data sets

The noisy threshold, Corollary S2 and the Markov rule need no numerical optimization, so `experiments/e36_analytic.rel` reran them on 2000 fresh data sets for each of the seven laws of Table 1. Every rule reached conditional coverage 0.9 in all 2000 data sets for every law, so each reliability has Clopper–Pearson 95% interval $[0.998, 1]$, including the t_3 law, outside $\mathcal{B}$. Mean conditional coverage was 0.969–0.992 for the noisy threshold, 0.968–0.991 for Corollary S2 and 0.984–0.998 for the Markov rule, and the 5% quantile of conditional coverage was at least 0.947, 0.945 and 0.968. The guarantees are therefore conservative in these designs: the latent laws are far from the extremal laws of § 3.

A stress test at the extremal law

The laws of Table 1 are far from extremal, so `experiments/e38_stress.py` uses the boundary-layer law $W = \theta - x^{1/2}E_u$ with $u = 0.0508$, the minimizer of the limiting latent coverage $e^{-ub_u(q)}$ at $q = 0.9$ (not the radius maximizer u^* of § S3), $x = 10^{-4}$, and θ solving the population noisy coverage $\text{pr}(|W + e| \leq 1) = 0.9$, with equal variances or with the lognormal variances of Table 1 scaled to mean x . The population latent coverage is then 0.89918 with equal variances, within 10^{-4} of the certified bound 0.8991 of Corollary 1, so the worst case at this noise level lies in $[0.8991, 0.89918]$, and 0.89959 with unequal ones. Conditional latent coverage is exact; $\delta = 0.05$; 20000 data sets for $K = 110$ and 1000 and 10000 for $K = 10^4$.

Table S1: Stress test, equal variances: mean conditional latent coverage and reliability (standard errors in parentheses), target 0.9 and 0.95

K	rank	k	mean latent coverage	reliability
110	marginal	100	0.9004 (0.0002)	0.531 (0.004)
	usual and certified	105	0.9503 (0.0002)	0.966 (0.001)
1000	marginal	901	0.89931 (0.00007)	0.482 (0.004)
	usual	916	0.91483 (0.00007)	0.943 (0.002)
	certified	917	0.91588 (0.00007)	0.954 (0.001)
10^4	marginal	9001	0.89918 (0.00003)	0.395 (0.005)
	usual	9050	0.90419 (0.00003)	0.922 (0.003)
	certified	9060	0.90522 (0.00003)	0.961 (0.002)

The usual rank keeps its noisy guarantee, with noisy reliability 0.952 and 0.954 at $K = 1000$ and 10^4 , but loses the latent one, by 4 and 10 standard errors; the certified rank of Proposition 3 restores it. The marginal rank has mean latent coverage below 0.9 once $K \geq 1000$, by 0.0008 at $K = 10^4$, the deficit of Corollary 1. With unequal variances the reliabilities are 0.948 and 0.960 at $K = 1000$, and 0.936 and 0.968 at $K = 10^4$, for the usual and the certified rank. At $K = 110$ the two ranks coincide and the order-statistic level 0.9068 leaves ample room. These simulations agree with the exact reliability $\text{pr}\{\text{Bin}(K, H(r_q)) \leq k - 1\}$ of Proposition 4, with $H(r_q) = 0.90080$: 0.943 and 0.954 at $K = 1000$,

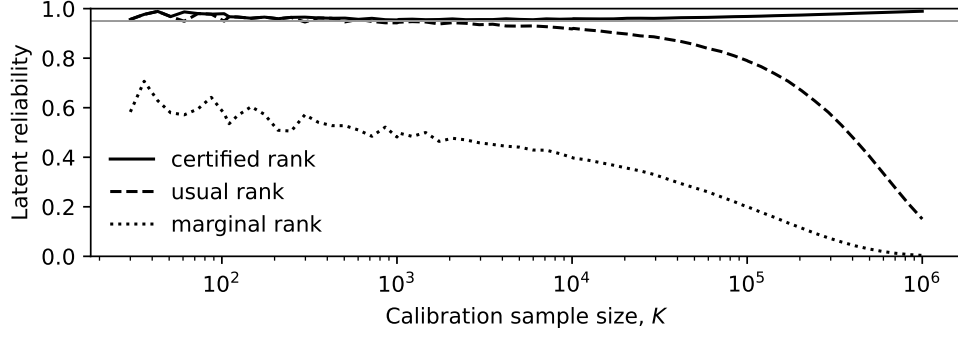


Figure S5: Exact latent reliability at the boundary-layer law of the stress test, equal variances with standard deviation 1% of the threshold, for the certified rank of Proposition 3 (solid), the usual high-probability rank (dashes) and the marginal rank (dots), $\delta = 0.05$; grey, the target 0.95.

Alt text: Line plot of latent reliability against calibration sample size on a logarithmic axis; the certified rank stays near or above 0.95 throughout, the usual rank declines from about 0.95 to 0.15 at one million, and the marginal rank declines from about 0.55 to near zero.

0.918 and 0.958 at $K = 10^4$ for the usual and the certified rank. Figure S5 extends the exact values to $K = 10^6$ (experiments/e40_exact_reliability.py); the usual rank falls to 0.790 at $K = 10^5$ and 0.151 at $K = 10^6$, the marginal rank to 0.004, while the certified rank stays above 0.95.

So the slack is not an artefact of the proofs: without it the latent guarantee fails at this law once the calibration sample is large enough for the order statistic to resolve a coverage difference of 10^{-3} . The effect needs small noise and a latent law with a sharp edge just beyond the threshold, here 2% beyond it; for the laws of Table 1 the guarantees are conservative.

Sensitivity to a shared latent law

Proposition 3 assumes that the calibration residuals and the new one share a law G , independent of the noise variances. experiments/e39_sensitivity.py uses the design of Table 1 with 2000 data sets for the normal, centred gamma and truncated exponential laws and two kinds of departure: residual standard deviation proportional to $D_i^{1/2}$ or $D_i^{-1/2}$, standardized to variance one, with the new area drawing its variance from the same population; and a new area whose residual is shifted by ± 0.2 or ± 0.5 latent standard deviations.

1. Averaged over the new area's variance, the dispersion departures left the reliability of the noisy threshold and of Corollary S2 at 1, with mean coverage 0.967–0.976, against 0.968–0.984 without departure. Conditionally on the new area's variance this fails: at the 10% quantile of the variances, when dispersion falls with the variance, and at the 90% quantile, when it rises, mean coverage was 0.918–0.942 and reliability as low as 0.75. Neither case is covered by Proposition 3, which assumes one law independent of the variances: averaged over new areas the reliability was kept in these designs, but not given a new area's variance.
2. Shifts of ± 0.2 changed mean coverage by at most 0.010, with reliability 1; shifts of ± 0.5 lowered it by up to 0.027, to 0.941 under the truncated exponential law, with reliability at least 0.977 and 5% quantile of conditional coverage 0.908.
3. At the boundary-layer law of the stress test, with $K = 110$ and 20000 data sets, the noisy threshold has reliability 0.969; shifting the new residual towards the edge by 0.1 latent standard deviations, 2%

of the threshold, lowered mean coverage to 0.860 and reliability to 0.035, and by 0.25 to 0.741 and 0, while shifts away from the edge raised coverage above 0.999.

The robustness in the first two items comes from the distance of these laws from the extremal ones, not from the margin between p_k and q ; near an extremal law a systematic difference between sampled and unsampled areas is not tolerated.

Oracle-matched widths

In a larger experiment with seven latent shapes, common and heterogeneous noise and 2000 data sets per setting, each rule’s width was rescaled by a single factor, chosen with knowledge of the latent law, to reach reliability 0.95 exactly. At matched reliability, marginal noisy conformal calibration was shortest in 12 of the 14 settings and the Fay–Herriot tolerance interval in the other 2; relative to noisy conformal calibration the Fay–Herriot tolerance interval was 3.1% shorter to 12.1% longer and the log-concave rule 4.5–11.5% longer (`results/conditional_synth_exact_matched.csv`). The log-concave rule with its guarantee was a median 15% (range 9–33%) wider than its own oracle-matched width, and the Fay–Herriot tolerance interval a median 6% (range –3 to 16%). So the advantage of the log-concave rule over noisy conformal calibration lies in obtaining a latent guarantee cheaply, not in a better interval shape.

Comparison with LatentCP and deconvolution

`experiments/e34_competitors.py` runs the following rules on the 600 data sets of the first four laws of Table 1; widths are relative to the computed radius. For the other three laws only the two analytic rules were run.

1. LatentCP (Zheng et al., 2026), single level with $\gamma = \alpha/2$, has a marginal guarantee. Its latent set depends on the new unit’s noise variance, which an unsampled area does not have; we draw it from the distribution that generated the D_i , 400 draws per data set, and average coverage and width over the draws. It was 16–18% longer, with mean coverage 0.976–0.990 and reliability 1.0. The tuned single-level and two-level versions, tuned on half the data, were 15–25% longer. A PAC version with a binomial rank, our adaptation, was 39–58% longer.
2. The deconvolution threshold of Cohen et al. (2025), with the mixture kernel in place of their homoscedastic kernel, collapses at $K = 110$ with their bin width 0.01: most bins are empty and the masked fit gives mean coverage 0.32–0.44. With bin width 0.2 and the empty bins fitted as zeros it had mean coverage 0.88–0.90, reliability 0.45–0.67 and width 22–30% shorter; it has no latent guarantee.
3. The noisy threshold of Proposition 3 was 10.5–11.7% longer and Corollary S2 9.1–10.2% longer, both with reliability 1.0.

The tuning grids of LatentCP and the scan step 0.01 of the deconvolution threshold are our choices; the original papers do not fix them for this setting.

School districts

In the finite population `apipop` (Lumley, 2004), 110 of 325 school districts are used for training and 110 for calibration, with 2 or 3 schools sampled per district. Coverage is the share of the remaining 105 districts that are covered. Design variances are estimated with 1–2 degrees of freedom and plugged in as known by every rule, a case that no guarantee covers. Over four conditions of 120 replications, the reliability of the heterogeneous rule was 0.942–0.992, of the mean-variance rule 0.983–1.000 and of the Fay–Herriot interval

0.908–0.992, below target in two conditions. The heterogeneous rule at level p_k was 0.8–5.2% shorter than the mean-variance rule with the $p = q$ table, mostly because of the level, and 10–16% longer than the Fay–Herriot interval.

S8 Reproducibility

All computations are in the public repository <https://github.com/kunhailP/latent-coverage-noisy-thresholds>; the submitted version is the tag `biometrika-submission`. The core routines are in `src/uai`: closed forms and the transition coefficient in `extremal.py`, the box certificates in `certify.py`, ball arithmetic in `interval.py`, and the procedures in `procedures.py` and `estimated.py`. The test suite, run by `make test`, checks the constants, the counterexamples and the centring bounds, and the `Makefile` lists the command that produces each group of results. The experiments are numbered scripts in `experiments`; `e27` produces the ball-arithmetic certificates, `e24` and `e25` the certified tables, `e20` Table 1, `e29` and `e30` the estimated-variance results, `e31` the shape-free rule, `e32` the centring and transition values, `e34` the comparison with LatentCP and deconvolution, `e35` and `fig_mean_location.py` Fig. S3, `fig_transition.py` Fig. S2, `e37` the coverage form of Corollary 1, `e38` the stress test, `e39` the sensitivity analysis and `e40` the exact reliability of Proposition 4 and Fig. S5; `uai.interval.bimodal.blc_certificate` checks Lemma S1. The analytic rules of Proposition 3 and Corollary S2 are `noisy_threshold_halfwidth` and `simple_shrink_halfwidth` in `procedures.py`; `hetldc_certified` gives the computed radius, while `hetldc_halfwidth` returns a grid value, a lower bound of R^{mix} , and is not a guaranteed radius.

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